Assignmnet 4

Berkeley Algorithm Viva Questions and Answers

1. What is Berkeley Algorithm?

Berkeley Algorithm is a clock synchronization algorithm used in distributed systems.

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2. What is the main aim of Berkeley Algorithm?

To synchronize clocks of all machines in a distributed network.

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3. What is clock synchronization?

Clock synchronization means maintaining the same time across all systems in a network.

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4. What is a distributed system?

A distributed system is a group of computers working together over a network.

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5. What is the role of the master node?

The master node collects time from slave nodes, calculates average time, and synchronizes clocks.

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6. What are slave nodes?

Slave nodes are client machines whose clocks are synchronized with the master clock.

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7. How is the master node selected?

The master node is selected using a leader election process.

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8. What happens in Berkeley Algorithm?

1. Master requests time from slaves
2. Slaves send their local time
3. Master calculates average difference
4. Updated time is broadcasted

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9. What is the purpose of average time calculation?

It reduces clock drift and improves synchronization accuracy.

10. What is clock drift?

Clock drift is the difference in time between system clocks.

11. Why are outliers ignored?

To improve synchronization accuracy by removing incorrect time values.

12. What is Cristian’s Algorithm?

Cristian’s Algorithm is another clock synchronization method using a time server.

13. How is Berkeley Algorithm better than Cristian’s Algorithm?

It improves accuracy by calculating average clock differences and ignoring outliers.

14. What is the role of sockets in this program?

Sockets are used for communication between server and clients.

15. Why is multithreading used in the program?

To handle multiple client connections simultaneously.

16. Which Python module is used for threading?

import threading

17. Which module is used for socket programming?

import socket

18. What is the purpose of datetime module?

It handles date and time operations.

19. What is the default port used in the program?

8080

20. What is startReceivingClockTime()?

It receives clock time from connected clients.

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21. What is synchronizeAllClocks()?

It synchronizes all client clocks using average clock difference.

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22. What is socket.bind()?

It binds the server to a specific port.

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23. What is socket.listen()?

It allows the server to accept incoming client connections.

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24. What is socket.connect()?

It connects the client to the server.

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25. Why is fault tolerance important in Berkeley Algorithm?

It helps the system continue working even if the master node fails.

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26. What are the features of Berkeley Algorithm?

- Centralized coordinator
- Clock adjustment
- Average calculation
- Fault tolerance
- Scalability
- Security

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27. What is scalability?

Scalability means handling many client systems efficiently.

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28. What is the purpose of time.sleep()?

It pauses execution for a specified time.

29. What is the final result of this practical?

Maintaining global synchronized time across all client machines.

30. What is the outcome of this practical?

Improvement in synchronization accuracy.